EXHIBIT 2

POSTMORTEM FORENSIC MEDICAL EXAMINATION REPORT
St. Tammany Parish Coroner's Office

EXHIBIT
Boyer 3
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Harmony Moller
36WF
Date of Autopsy: 02/08/2023 at 10:45 A.M.

FINAL DIAGNOSIS:
1.0 Abnormally high left coronary ostia takeoff exacerbated by mitragynine use
2.0 Pulmonary edema and congestion
3.0 Chronic passive congestion of liver
4.0 Mild nephrosclerosis

CAUSE OF DEATH: Abnormally high left coronary ostia takeoff exacerbated by mitragynine use

MANNER OF DEATH: Accident

EXTERNAL DESCRIPTION

The body bag is sealed with a tag labeled "136846". The body is received nude from LOPA. Clothing is noted in a plastic bag within the body bag. Property removed from the decedent consists of two yellow metal rings with a clear stone from the right hand.

The body is that of a well-developed, well-nourished, 68" white female whose appearance is consistent with the given age of 36 years. The weight of the body is 152 pounds. The scalp is covered with long, brown hair. There is marked rigidity in the extremities. Lividity is noted on the posterior surfaces of the body. There are no noticeable tattoos or scars. The corneas have been harvested prior to autopsy. The ears and nose are patent and unremarkable. The teeth are natural and in good condition. The lips and anterior oral mucosa are free of trauma. The chest is sunken and the abdomen is flat. The back is straight. The upper and lower extremities are present.

Evidence of Therapy:
There is a curvilinear sutured incision on the chest. There are sutured incisions down the insides of both legs consistent with long bone and heart valve recovery. An endotracheal tube is in the mouth and correctly placed in the trachea. A venipuncture mark is in the right anterior neck. There is a sutured defect on the posterior left leg.

Evidence of Injury: None

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INTERNAL EXAMINATION

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Examination of the body cavities reveals minimal amounts of fluid in the pleural and abdominal cavities. The pericardial sac is open.

Neck Organs: The tongue, hyoid bone, thyroid gland, thyroid cartilage, larynx, and muscles of the anterior neck have no evidence of disease or injury.

Heart and Great Vessels: Not present.

Lungs: The 720 gram right lung and 480 gram left lung have marked subpleural anthracotic pigmentation. The pulmonary vessels are patent and no emboli are present. The bronchi contain blood. Pulmonary cut surfaces demonstrate hemorrhage, congestion, and edema. There is no evidence of tumor, inflammatory consolidation, or granulomatous disease.

Liver and Gallbladder: The 1940 gram liver has a smooth, glistening, intact capsule. The gallbladder contains dark, green bile and is free of stones. The lymph nodes at the porta hepatis are not enlarged. The cut hepatic surfaces contain dark brown and light tan areas, along with areas of a nutmeg pattern consistent with chronic passive congestion.

Spleen: The 150 gram spleen has a smooth, glistening, intact, thin, slate gray-purple capsule. The dark maroon, granular cut surfaces do not have a prominent follicular or trabecular architecture.

Adrenals: The adrenals are symmetrical and have unremarkable external and cut features.

Pancreas: The pancreas lies in its normal retroperitoneal location and on cut surface demonstrates a normal lobulated parenchyma.

Kidneys: The 130 gram right kidney and the 170 gram left kidney have granular subcapsular surfaces. The cut surfaces are dark, red-brown with normal corticomedullary ratios and clearly defined demarcations. The ureters are thin-walled and not dilated. The bladder contains clear, yellow urine.

Gastrointestinal Tract: The small and large bowels follow their normal anatomic course and are unremarkable along their serosal surfaces. The appendix is present. The stomach contains an estimated 25 cc of brown fluid material. The mucosa of the duodenum, esophagus, and stomach is unremarkable.

Musculoskeletal: The sternal plate has been removed secondary to LOPA recovery. There are no fractures of the clavicles, ribs, or vertebral column. The subcutaneous tissues of the chest and abdomen are free of hemorrhage and trauma.

Reproductive Organs: The uterus, fallopian tubes, and ovaries are present and grossly unremarkable externally and on cut surfaces. There is no evidence of intrauterine pregnancy.

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Brain: The scalp and underlying subcutaneous tissues are free of any areas of hemorrhage or trauma. The cerebrospinal fluid surrounding the 1230 gram brain is clear, colorless, and normal in quantity. The vessels at the base of the brain show minimal degrees of atherosclerotic narrowing. The cerebral hemispheres are symmetrical and have unremarkable external surfaces. Cut surfaces of the cerebrum, midbrain, pons, cerebellum, medulla, and medulla oblongata are unremarkable. After stripping the dura, there are no underlying skull fractures.

Specimens for Laboratory Studies: Blood, vitreous, urine, and liver specimens are retained for drug and alcohol screen.

Michael DeFatta MD

Michael DeFatta, M.D., Forensic Pathologist

6-6-23

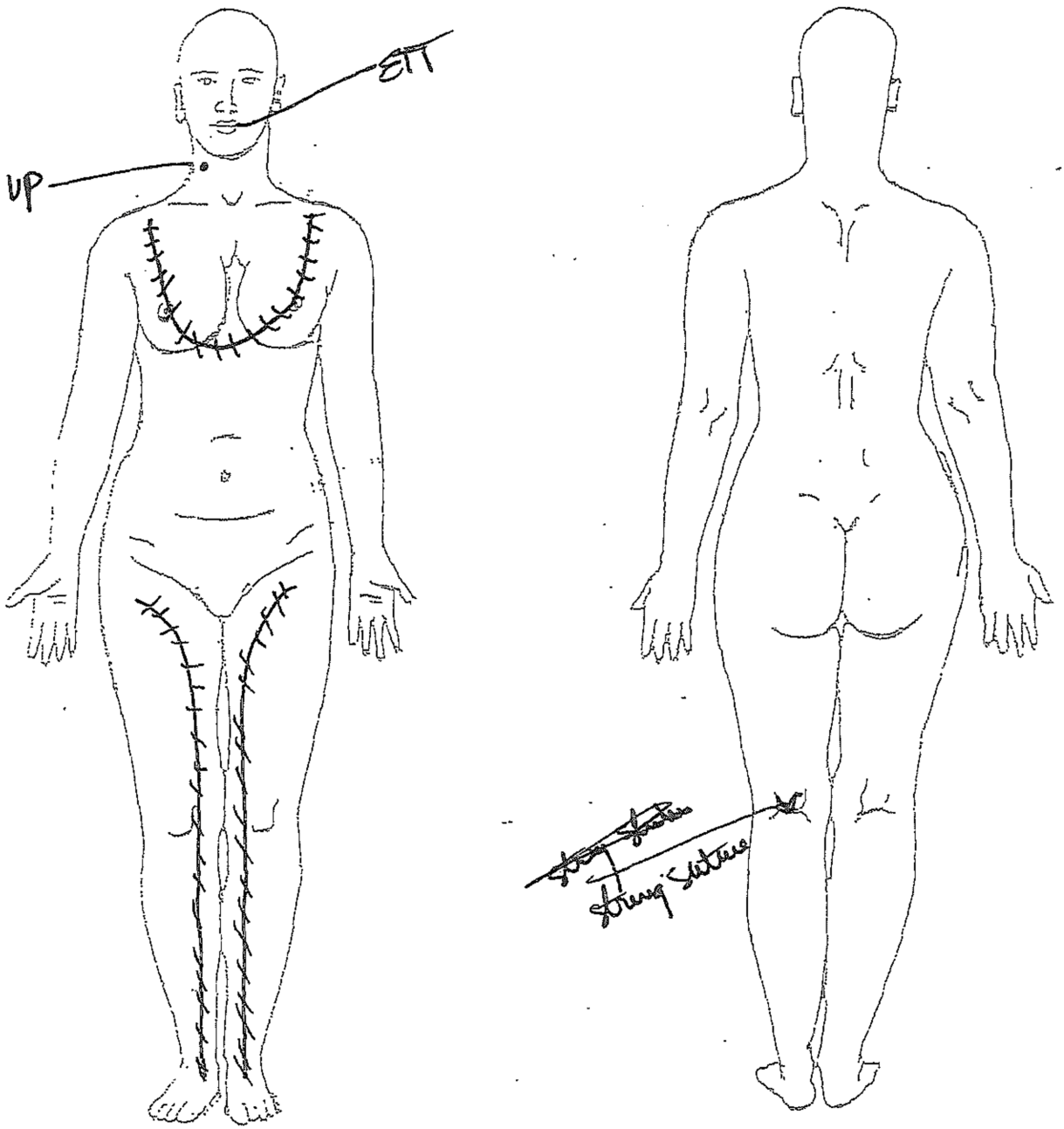
Date

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Case # ST-1147-23 Name Harmony Montgomery

Age/Sex/Race 36/WF Height _____ Weight 165



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Case 2:24-cv-00228-WBV-DPC Document 249-3 Filed 03/02/26 Page 5 of 11

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Robert A. Middleberg, PhD, F-ABFT, DABCC-TC, Laboratory Director

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Toxicology Report

Report Issued 02/24/2023 12:06

To: 147883
St. Tammany Parish Coroner (LDH OSP)
Attn: Ana Welch
65278 Hwy 434
Lacombe, LA 70445

Patient Name	Moller, Harmony
Patient ID	230206-995
Chain	230206-995
DOB	02/18/1986
Sex	Female
Workorder	23055002

Positive Findings:

Analyte	Result	Units	Matrix Source
Caffeine	Presump Pos	mcg/mL	001 - Femoral Blood
Naloxone	Presump Pos	ng/mL	001 - Femoral Blood
Delta-9 Carboxy THC	10	ng/mL	001 - Femoral Blood
Delta-9 THC	8.6	ng/mL	001 - Femoral Blood
Diphenhydramine	64	ng/mL	001 - Femoral Blood
Promethazine	18	ng/mL	001 - Femoral Blood
Mitragynine	150	ng/mL	001 - Femoral Blood
Creatinine (Vitreous Fluid)	0.864	mg/dL	003 - Vitreous Fluid
Sodium (Vitreous Fluid)	142	mmol/L	003 - Vitreous Fluid
Potassium (Vitreous Fluid)	7.47	mmol/L	003 - Vitreous Fluid
Chloride (Vitreous Fluid)	118	mmol/L	003 - Vitreous Fluid
Glucose (Vitreous Fluid)	18.2	mg/dL	003 - Vitreous Fluid
Urea Nitrogen (Vitreous Fluid)	17.2	mg/dL	003 - Vitreous Fluid

See Detailed Findings section for additional information

Agency Case Number: ST-1147-23

Testing Requested:

Test	Test Name
1919FL	Electrolytes and Glucose Panel (Vitreous), Fluid (Forensic)
8042B	Postmortem, Expanded w/Vitreous Alcohol Confirmation, Blood (Forensic)

Specimens Received:

ID	Tube/Container	Volume/ Mass	Collection Date/Time	Matrix Source	Labeled As
001	Gray Stopper Plastic Tube	3 mL	02/07/2023 11:39	Femoral Blood	MOLLER, H.
002	Gray Stopper Plastic Tube	3 mL	02/07/2023 11:39	Femoral Blood	MOLLER, H.
003	Red Stopper Plastic Tube	1.75 mL	02/07/2023 11:39	Vitreous Fluid	HARMONY MOLLER
004	Red Stopper Plastic Tube	1.75 mL	02/07/2023 11:39	Vitreous Fluid	HARMONY MOLLER
005	Red Stopper Plastic Tube	2 mL	02/07/2023 11:39	Vitreous Fluid	KENNETH STEGALL
006	Red Stopper Plastic Tube	1.25 mL	02/07/2023 11:39	Vitreous Fluid	KENNETH STEGALL
007	White Cap Plastic Container	13.2 g	02/07/2023 11:39	Liver Tissue	230206-995

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Workorder 23055002
Chain 230206-995
Patient ID 230206-995

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All sample volumes/weights are approximations.
Specimens received on 02/10/2023.

Detailed Findings:

Analysis and Comments	Result	Units	Rpt. Limit	Specimen Source	Analysis By
Caffeine	Presump Pos	mcg/mL	0.20	001 - Femoral Blood	LC/TOF-MS
This test is an unconfirmed screen. Confirmation by a more definitive technique such as GC/MS is recommended.					
Naloxone	Presump Pos	ng/mL	1.0	001 - Femoral Blood	LC/TOF-MS
This test is an unconfirmed screen. Confirmation by a more definitive technique such as GC/MS is recommended.					
Delta-9 Carboxy THC	10	ng/mL	5.0	001 - Femoral Blood	LC-MS/MS
Delta-9 THC	8.6	ng/mL	0.50	001 - Femoral Blood	LC-MS/MS
Diphenhydramine	64	ng/mL	50	001 - Femoral Blood	LC-MS/MS
Promethazine	18	ng/mL	5.0	001 - Femoral Blood	LC-MS/MS
Substance(s) known to interfere with the identity and/or quantity of the reported result: Promazine; Chlorpromazine					
Mitragynine	150	ng/mL	50	001 - Femoral Blood	LC-MS/MS
Creatinine (Vitreous Fluid)	0.864	mg/dL	0.500	003 - Vitreous Fluid	Chemistry Analyzer
Sodium (Vitreous Fluid)	142	mmol/L	50.0	003 - Vitreous Fluid	Chemistry Analyzer
Potassium (Vitreous Fluid)	7.47	mmol/L	1.00	003 - Vitreous Fluid	Chemistry Analyzer
Chloride (Vitreous Fluid)	118	mmol/L	50.0	003 - Vitreous Fluid	Chemistry Analyzer
Glucose (Vitreous Fluid)	18.2	mg/dL	10.0	003 - Vitreous Fluid	Chemistry Analyzer
Urea Nitrogen (Vitreous Fluid)	17.2	mg/dL	2.00	003 - Vitreous Fluid	Chemistry Analyzer

Other than the above findings, examination of the specimen(s) submitted did not reveal any positive findings of toxicological significance by procedures outlined in the accompanying Analysis Summary.

Reference Comments:

1. Chloride (Vitreous Fluid) (Cl-) - Vitreous Fluid:
   Normal: 105 - 135 mmol/L
2. Creatinine (Vitreous Fluid) - Vitreous Fluid:
   Normal: 0.6 - 1.3 mg/dL
3. Delta-9 Carboxy THC (Inactive Metabolite) - Femoral Blood:
   Delta-9 THC is the principle psychoactive ingredient of marijuana/hashish. Delta-9 carboxy THC (THCC) is the inactive metabolite of THC. The usual peak concentrations in serum for 1.75% or 3.55% THC marijuana cigarettes are 10-101 ng/mL attained 32 to 240 minutes after beginning smoking, with a slow decline thereafter. The ratio of whole blood concentration to plasma concentration is unknown for this analyte. THCC may be detected for up to one day or more in blood. Both delta-9-THC and THCC may be present substantially longer in chronic users. THCC is usually not detectable after passive inhalation.

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Workorder	23055002
Chain	230206-995
Patient ID	230206-995

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Reference Comments:

4. Delta-9 THC (Active Ingredient of Marijuana) - Femoral Blood:

   Delta-9 THC is the principle psychoactive ingredient of marijuana (cannabis, hashish). It is also the active component of the prescription medication Marinol®. Marijuana use causes relaxation, distorted perception, euphoria and feelings of well being, along with confusion, dizziness, somnolence, ataxia, speech difficulties, lethargy and muscular weakness.

   After smoking a user-preferred 300 mcg/kg dose average plasma THC concentrations at 35 minutes were reported at 16.1 (range 4.7-30.9) ng/mL, and had declined to 1.5 (range 0.4-3.2) ng/mL after 190 minutes. Usual peak levels in serum for 1.75% or 3.55% THC marijuana cigarettes: 50-270 ng/mL at 6 to 9 minutes after beginning smoking, decreasing to less than 5 ng/mL by 2 hrs. Whole blood THC concentrations are typically half those in a corresponding plasma sample.

5. Diphenhydramine (Benadryl®; Ingredient of Benylin and Panadol; Nytol; Unisom) - Femoral Blood:

   Diphenhydramine is an antihistamine with sedative and anti-emetic effects. It is rapidly absorbed following oral administration; however, it is frequently given IV. Patients taking this medication are usually warned against the operation of complicated machinery, because of its strong sedative effects.
   Following a single 50 mg oral dose of diphenhydramine, peak plasma concentrations at 2.3 hr averaged 66 ng/mL.
   Signs and symptoms of acute diphenhydramine toxicity include tremor, seizures, fever, respiratory depression and cardiac arrhythmias. The average blood diphenhydramine concentrations reported in fatal overdoses were 1400 ng/mL in infants, 4400 ng/mL in children and 15000 ng/mL in adults.
   The blood to plasma concentration ratio for diphenhydramine is approximately 0.80.

6. Glucose (Vitreous Fluid) (C6H12O6; D-glucose (biologically active); Dextrose; L-glucose) - Vitreous Fluid:

   Normal: <200 mg/dL

   Postmortem vitreous glucose concentrations >200 mg/dL are associated with hyperglycemia.

   Since postmortem vitreous glucose concentrations decline rapidly after death both in vivo and in vitro, care should be taken in the interpretation of results. Stability of vitreous glucose for up to 30 days has been noted by NMS Labs when specimens are maintained frozen (-20C).

7. Mitragynine (Kratom) - Femoral Blood:

   Mitragynine is an alkaloid found in the plant Kratom which originates from Asia and is considered an atypical opioid agonist. The leaves of the plant are traditionally chewed or brewed into a tea, however, Kratom powders and leaf extracts are also sold and may be subject to abuse. The effects of Kratom are dose dependent, with lower doses resulting in stimulant-type effects and higher doses resulting in opioid-type effects. Reported adverse effects include seizures, psychosis, and liver toxicity.

   Over a 27 month-period, mitragynine was reported in 1001 blood specimens submitted during the course of medicolegal investigations; concentrations ranged from 5.6-29,000 ng/mL. In three cases which ruled mitragynine as the sole drug in the cause of death, postmortem peripheral blood concentrations ranged from 1590-3420 ng/mL.

8. Potassium (Vitreous Fluid) (K+) - Vitreous Fluid:

   Normal: <15 mmol/L
   Quantitative results for Potassium will be affected if performed on gray top tubes since these collection tubes contain potassium oxalate.

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Workorder 23055002
Chain 230206-995
Patient ID 230206-995

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Reference Comments:

9. Promethazine (Phenergan®) - Femoral Blood:

   Promethazine is an ethylamino-derivative of phenothiazine used for its antihistaminic, antiemetic and sedative effects. It generally produces CNS depression at the usual therapeutic range; however, promethazine also can produce CNS stimulation or it may be added to other substances and used in the treatment of allergies and management of motion sickness. Oral doses usually range from 25 to 150 mg per day.

   Following a single 50 mg oral administration of promethazine in 20 younger men, the average peak serum concentration of 29 ng/mL (range 6-99) occurred at 3.3 hours. Promethazine has a blood / plasma ratio of 0.6-0.7.

   Postmortem blood concentrations of promethazine in fatalities due to promethazine overdose were reported to be between 1800 and 12000 ng/mL. Promethazine may exhibit postmortem redistribution.

10. Sodium (Vitreous Fluid) (Na+) - Vitreous Fluid:

    Normal: 135 - 150 mmol/L
    Quantitative results for sodium will be affected if performed on gray top tubes since these collection tubes contain sodium fluoride.

11. Urea Nitrogen (Vitreous Fluid) (Carbamide; carbonyl diamide; carbonyldiamine) - Vitreous Fluid:

    Normal: 8 - 20 mg/dL

Sample Comments:

001 Physician/Pathologist Name: Michael DeFatta, M.D.

Unless alternate arrangements are made by you, the remainder of the submitted specimens will be discarded one (1) year from the date of this report; and generated data will be discarded five (5) years from the date the analyses were performed.

Workorder 23055002 was electronically signed on 02/24/2023 11:57 by:

Andrea Carolina Noble, Ph.D.
Forensic Toxicologist

Analysis Summary and Reporting Limits:

All of the following tests were performed for this case. For each test, the compounds listed were included in the scope. The Reporting Limit listed for each compound represents the lowest concentration of the compound that will be reported as being positive. If the compound is listed as None Detected, it is not present above the Reporting Limit. Please refer to the Positive Findings section of the report for those compounds that were identified as being present.

Test 1919FL - Electrolytes and Glucose Panel (Vitreous), Fluid (Forensic) - Vitreous Fluid

-Analysis by Chemistry Analyzer for:

Analyte	Rpt. Limit	Analyte	Rpt. Limit
Chloride (Vitreous Fluid)	50.0 mmol/L	Potassium (Vitreous Fluid)	1.00 mmol/L
Creatinine (Vitreous Fluid)	0.500 mg/dL	Sodium (Vitreous Fluid)	50.0 mmol/L
Glucose (Vitreous Fluid)	10.0 mg/dL	Urea Nitrogen (Vitreous Fluid)	2.00 mg/dL

Test 52198B - Cannabinoids Confirmation, Blood - Femoral Blood

-Analysis by High Performance Liquid Chromatography/ Tandem Mass Spectrometry (LC-MS/MS) for:

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Workorder 23055002
Chain 230206-995
Patient ID 230206-995

Analysis Summary and Reporting Limits:

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Analyte	Rpt. Limit	Analyte	Rpt. Limit
11-Hydroxy Delta-9 THC	1.0 ng/mL	Delta-9 THC	0.50 ng/mL
Delta-9 Carboxy THC	5.0 ng/mL		

Test 52441B - Diphenhydramine Confirmation, Blood - Femoral Blood

-Analysis by High Performance Liquid Chromatography/ Tandem Mass Spectrometry (LC-MS/MS) for:

Analyte	Rpt. Limit	Analyte	Rpt. Limit
Diphenhydramine	50 ng/mL		

Test 52456B - Promethazine Confirmation, Blood - Femoral Blood

-Analysis by High Performance Liquid Chromatography/ Tandem Mass Spectrometry (LC-MS/MS) for:

Analyte	Rpt. Limit	Analyte	Rpt. Limit
Promethazine	5.0 ng/mL		

Test 52495B - Mitragynine Confirmation, Blood - Femoral Blood

-Analysis by High Performance Liquid Chromatography/ Tandem Mass Spectrometry (LC-MS/MS) for:

Analyte	Rpt. Limit	Analyte	Rpt. Limit
Mitragynine	50 ng/mL		

Test 8042B - Postmortem, Expanded w/Vitreous Alcohol Confirmation, Blood (Forensic) - Femoral Blood

-Analysis by Enzyme-Linked Immunosorbent Assay (ELISA) for:

Analyte	Rpt. Limit	Analyte	Rpt. Limit
Barbiturates	0.040 mcg/mL	Gabapentin	5.0 mcg/mL
Cannabinoids	10 ng/mL	Salicylates	120 mcg/mL

-Analysis by Headspace Gas Chromatography (GC) for:

Analyte	Rpt. Limit	Analyte	Rpt. Limit
Acetone	5.0 mg/dL	Isopropanol	5.0 mg/dL
Ethanol	10 mg/dL	Methanol	10 mg/dL

-Analysis by High Performance Liquid Chromatography/Time of Flight-Mass Spectrometry (LC/TOF-MS) for: The following is a general list of analyte classes included in this screen. The detection of any specific analyte is concentration-dependent. Note, not all known analytes in each specified analyte class are included. Some specific analytes outside of these classes are also included. For a detailed list of all analytes and reporting limits, please contact NMS Labs. Amphetamines, Anticonvulsants, Antidepressants, Antihistamines, Antipsychotics, Benzodiazepines, CNS Stimulants, Cocaine and Metabolites, Hallucinogens, Hypnosedatives, Muscle Relaxants, Non-Steroidal Anti-Inflammatory Agents, Opiates and Opioids.

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UCSF
Clinical Toxicology
and Environmental Biomonitoring Laboratory

UCSF Mt Zion Cancer
Research Building
2340 Sutter St.
Room S-271
San Francisco, CA 94115
Tel #: (415) 502-1446

SAMPLE ANALYSIS REPORT

Date: 04/18/2023
Date Samples Submitted: 04/14/2023
DEA Sample ID: DEA-2023-1183
Contract: Samples Analyzed for the Drug Enforcement Administration (DEA) (contract #: 15DDHQ19F0000086)
Sample Origin: St. Tammany Parrish Coroner, Lacombe, LA
Testing Requested: Comprehensive Drug Analysis
Analytical Platform: LC-Quadrupole Time-of-Flight Mass Spectrometer (Agilent LC1260- QTOF/MS 6550)

Findings:

Sample ID	Sample	Date	Confirmed Drug (ng/mL)
230207-LA-70445-001-WB1	Whole Blood	02/07/23	1. Mitragynine (158) 2. 7-OH Mitragynine* (0.8**) 3. Diphenhydramine 4. Naloxone 5. Promethazine

NOTES:

1. *7-OH mitragynine is an expected metabolite of mitragynine.

2. **Value was below the lower limit of quantification of 3ng/mL and equal to the limit of detection for the analysis.

3. Drug libraries used in analysis:
   Comprehensive Drug Library consisting of 912 New Psychoactive Substances, 161 Traditional Illicit Drugs, 27 Dietary Supplement Stimulants, and 93 Prescription Drugs or Over-the-Counter Drugs

Analyst:

Ross R Ellison

Ross Ellison,
Staff Research Associate

Principal Investigator:

Roy Gerona, Ph.D.

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11/17/23
St. Tammany Parish Coroner's Office
Harmony Moller